

The entire air distribution system shall be balanced using an anemometer. Measured air quantities at fan discharge and at various outlets shall be identical to or less/excess than 5 percent of those specified and quoted. Branch duct adjustments shall be permanently marked after air balancing is completed so that these can be restored to their correct position if disturbed at any time. Complete air balance report shall be submitted for scrutiny and approval, and four copies of the approved balance report shall be provided with completion documents.

43. INSULATION

43.1. Scope

The scope of this section comprises the supply and application of insulation conforming to these specifications. The insulation material shall be Closed Cell Elastomeric Nitrile Rubber / Polyethylene Foam / EPDM

43.2. Material

- Thermal insulation material for Duct & Pipe insulation shall be with factory laminated fiber glass cloth closed cell Elastomeric UV resistant. Thermal conductivity as per **BS 874 part 2 – 86 (DIN 52613 52612) / DIN EN 12667 / EN ISO8497** of the insulation material shall not exceed 0.038 W/m²K or 0.212 BTU / (Hr-ft²-°F/inch) at an average temperature of 30°C. Density of the nitrile rubber shall be 40-60 Kg/m³, for EPDM shall be 40-60 Kg/m³ & for polyethylene material it shall be 25-30 Kg/m³. The product shall have temperature range of –40 °C to 105°C. The insulation material shall be fire rated for Class 0 as per BS 476 Part 6 : 1989 for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test. Water vapour permeability shall be not less than 0.024 per inch (2.48 x 10⁻¹³ Kg/m.s.Pa i.e. $\mu \geq 7000$: Water vapour diffusion resistance) as per **DIN 53122 part 2, DIN 52615 / EN 12086 & EN13469**.

Comply

In addition to above properties the insulation material for ducts shall be anti-microbial. Microbiological growth on insulation surface shall be in accordance with ASTM G-21 and bacterial resistance to ASTM2180.

The Material shall comply to ISO 5659 / BS 6853 / ABD 0031 for smoke density and toxicity values. The thermal conductivity of insulation material shall not be effected by aging as per **DIN 52616 standard**.

Insulation shall be with self-adhesive for ducting and piping and available in roles / sheets.

Thickness of the insulation shall be as specified for the individual application. **Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for density and thickness.** Samples of insulation material from each lot delivered at site may be selected by Owner's site representative and gotten tested for thermal conductivity and density at Contractor's cost. Adhesive used for sealing the insulation shall be non-flammable and with low VOC content (maximum 850 gm/l as per IGBC guide lines) strictly as per manufacturer's recommendations.

Ducting insulation thickness shall be as per table below.

Ducting position	Thickness
SA duct in RA path	19 mm
Ducted return air system	SA duct: 25 mm RA duct: 19 mm
Both SA & RA exposed	Both 25 mm
TFA ductwork for datahall MAU	32mm
SA duct having open cell nitrile rubber insulation	9 mm

43.3. Duct Insulation

External thermal insulation shall be provided as follows :

The thickness of insulation material shall be as shown on drawings or identified in the schedule of quantity. Following procedure shall be adhered to:

Noted

Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work. Measurement of surface dimensions shall be taken properly to cut closed cell insulation to size with sufficient allowance in dimension. Cutting of insulation sheets shall be done with adjustable blade to make 90° cut in thickness of sheet. Hacksaw or blades are not acceptable tools for cutting the insulation.

Material shall be fitted under compression and no stretching of material shall be permitted. All longitudinal and transverse joints shall be sealed by providing 50 mm wide Fibre glass cloth laminated tape as per manufacturer recommendations. The insulation installers shall be certified by manufacture.

Where ducts/pipes penetrates walls / floor it shall be insulated with intumescent properties insulation material for fire protection. The treatment shall be minimum 500 mm extended on both sides.

43.4. Piping Insulation

All chilled water, refrigerant, and condensate drain piping shall be insulated in the manner specified herein. Before applying insulation, all pipe shall be brushed and cleaned. All MS pipes shall be provided with a coat of zinc chromate primer. Thermal insulation shall be applied as follows or as specified in drawings or schedule of quantity:

Piping Insulation thickness shall be as follows;

Pipe nominal bore	Piping on Terrace	Piping in Riser	Piping on Floor
	Nitrile rubber insulation with glass cloth and Aluminum cladding		Nitrile rubber insulation with glass cloth
700 mm dia (50mm thick insulation)	50mm thick	44mm	NA
650 mm dia (50mm thick insulation)	75mm thick	44mm	NA
600 mm dia (50mm thick insulation)	75mm thick	44mm	NA

Noted

Pipe nominal bore	Piping on Terrace	Piping in Riser	Piping on Floor
	Nitrile rubber insulation with glass cloth and Aluminum cladding		Nitrile rubber insulation with glass cloth
550 mm dia (50mm thick insulation)	50mm	44mm	NA
500 mm dia (50mm thick insulation)	50mm	44mm	NA
450 mm dia (50mm thick insulation)	50mm	44mm	NA
400 mm dia (50mm thick insulation)	50mm	44mm	NA
350 mm dia (41mm thick insulation)	41mm	44mm	32mm
300 mm dia (41mm thick insulation)	41mm	44mm	32mm
250 mm dia (32mm thick insulation)	32mm	44mm	25mm
200 mm dia (32mm thick insulation)	32mm	44mm	25mm
150 mm dia (32mm thick insulation)	32mm	44mm	25mm
125 mm dia (32mm thick insulation)	32mm	32mm	25mm
100 mm dia (32mm thick insulation)	32mm	32mm	25mm
80 mm dia (32mm thick insulation)	32mm	32mm	25mm
65 mm dia (32mm thick insulation)	32mm	32mm	19mm
50mm dia (25mm thick insulation)	25mm	25mm	19mm
40mm dia (25mm thick insulation)	25mm	25mm	19mm
32mm dia (25mm thick insulation)	25mm	25mm	19mm
25mm dia (25mm thick insulation)	25mm	25mm	19mm

Noted

Insulating material in tube form (minimum upto 100 dia pipes) shall be sleeved on the pipes. On piping, slit opened tube from insulating material shall be placed over the pipe and adhesive shall be applied as suggested by the manufacturer. Adhesive must be allowed to tack dry and then press surface firmly together starting from butt end and working towards centre. Wherever flat sheets shall be used it shall be with self-adhesive and cut out in correct dimension using correct tools. Scissors or Hacksaw-blade shall not be allowed. All longitudinal and transverse joints shall be sealed by providing 50 mm wide fiber glass cloth laminated tape as per manufacturer recommendations. The adhesive shall be strictly as

recommended by the manufacturer. The insulation shall be continuous over the entire run of piping, fittings and valves. All valves, fittings, joints, strainers etc. in chilled water piping shall be insulated to the same thickness as specified for the main run of piping and application shall be same as above. Valves bonnet, yokes and spindles shall be insulated in such a manner as not to cause damage to insulation when the valve is used or serviced.

Direct contact between pipe and hanger shall be avoided. Hangers shall pass outside the saddle. Manufacturer shall supply PUF saddles with pre-laminated insulation sheet of both side (PUF saddle sandwich between insulation material on both side) so that the insulation material is joint with insulation material on both side (only for Nitrile & EPDM) and the weight of pipe is transferred to the PUF saddle in the center.

Manufacturer's installation manual shall be submitted and followed for full compliance. All insulation work shall be carried out by skilled workmen specially trained and certified by manufacturer in this kind of work. All insulated pipes shall be labeled (S.R. or R.R.) and provided with 300 mm wide band of paint along circumference at every 1200 mm for colour coding. Direction of fluid shall also be marked. Un-insulated MS pipes shall be painted throughout and direction of fluid marked. All painting shall be as per relevant BIS codes.

43.5. Protective Coating / Vapour Barrier Over Insulation

All ducts and pipes (On the roof / outside) exposed to UV rays shall be covered with two coats of UV paint / epoxy.

43.6. Duct Acoustic Lining

Open Cell Nitrile Rubber

Duct acoustic lining material shall be Nitrile Rubber open cell foam. Thermal conductivity of the insulation material shall not exceed 0.047 W/m²K at an average temperature of 20°C. Density of the nitrile rubber shall be 140 – 180 Kg/m³. The material should withstand maximum surface temperature of +85°C and minimum surface temperature of -20°C. The material should conform to Class 1 rating for surface spread of Flame in accordance to BS 476 Part 7 & HBF, HF 1 & HF 2 in accordance to UL 94, 1996.

Comply

Insulation should have antimicrobial product protection, and should pass Fungi Resistance as per ASTM G 21 and Bacterial Resistance as per ASTM E 2180. The insulation should pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071-05 (section 12.7).

Thickness of the material shall be 15 mm thick specified for the individual application and with noise absorption proprieties as per IS: 8225 / ISO 354 / ASTM423C. The insulation should be installed as per manufacturer's recommendation.

43.7. Data Centre Floor Insulation

Floor of data centre shall be insulated with 19mm thick aluminum foil faced nitrile rubber / XLPE insulation as per specifications. 600mm x 600mm grid shall be made on the floor. (Insulation grid shall be of same size as of data centre floor grid). This grid shall be made from 19 mm wide 22G GI C-section. These C-sections shall be screwed to floor. Depth of screwing in to the floor shall not be more than 5mm. Sections of insulation shall be stuck to floor with self adhesive / adhesive solution as recommended by insulation manufacturer.

43.8. Underdeck Insulation For Roof

Underdeck insulation shall be either of the following with density greater than or equal to 25 Kg/m³ & thermal conductivity of 0.21 Btu in/ft² hr° F (at 24° C as per ASTM C-518).

- i. 50mm thick fiberglass slabs with aluminium foil both sides
- ii. 50mm thick Closed cell elastomeric nitrile rubber